



Investigation report

Haul truck and light vehicle collision, Bindarra Gold Mine - NTSB-style probable cause report

Drafted 04/10/2026

INVESTIGATION WORKFLOW SUITE

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Executive summary

What happened

At 02:47 on 12/08/2026, an empty 220 t class rigid haul truck, HT-407, was descending the main ramp of the Stage 4 open pit at Bindarra Gold Mine when a light vehicle, LV-118, emerged from the RL 312 dewatering spur into its path. The truck's front left struck the light vehicle's right side at 24 km/h, pushed it 18.4 m down the ramp and rolled it onto its left side against the inner windrow. The driver, a grade control technician employed by a survey subcontractor, was on his third night shift at the mine. He remained conscious, was extricated through the rear of the vehicle 27 minutes after the collision, and was flown to the metropolitan trauma centre at 07:40.

He sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. His seatbelt was fastened and showed loading marks. The roll-over protection structure deformed inward without being breached. The light vehicle was destroyed. The truck, damaged at the front left, returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes and the main ramp below RL 340 was closed for 22 hours. The occurrence was notified to the State Mines Inspectorate as a reportable injury.

What the investigation found

The probable cause was that the RL 312 spur was opened to traffic under a works order with no management of change raised. The resulting intersection, with 41 m of sight line against a 96 m stopping distance, was unsigned, unlisted for positive radio calls, and absent from the in-cab road network and the collision awareness system configuration. The light vehicle entered the ramp without stopping or establishing the truck's position. It is very likely that neither operator was aware of the other until about four seconds before the collision.

The works order exclusion at CRL-PRO-022 clause 5.3 removed the three referrals that would have reached the traffic management plan, the in-cab map and the geofence configuration, which very likely contributed. The spur met the ramp 34 m beyond a crest, well short of the plan minimum of 120 m, which very likely contributed. BIN-TMP-004 required positive calls only at nine listed intersections, with the list reissued yearly, which very likely contributed. Neither operator made a call, which likely contributed. The S4-LOAD suppression geofence, extended on 09/06/2026, covered the intersection. This likely contributed through the operator's expectation that silence on that ramp meant an empty road, although the single suppressed alert would have come after braking began. The driver's contractor induction was accepted without the site supplement or pit driving assessment being verified, which likely contributed. The discretion left by MSG-14 likely contributed, although this finding is less firm. The mine was not in breach.

The truck operator braked and steered towards the inner windrow 1.4 seconds after the last steady data point, reducing the impact speed from 37 km/h to 24 km/h.

The single paramedic night roster and the operator's position on night 10 of a 14 night swing did not contribute, but each increased risk. A wet ramp and degraded brakes were ruled out. Whether dust reduced visibility could not be determined. A water cart pass was overdue and particulate was elevated, but the driver described clear air, the sighting matched clear-air geometry, and nothing measured visibility.

What can be done

A road cut into an operating pit changes the traffic system, and every control that depends on knowing where intersections are must be told.

At 05:10 on 12/08/2026, the Mining Manager required positive calls at every Stage 4 intersection until BIN-TMP-004 is amended. The spur was barricaded and remains closed. On 14/08/2026, survey measured the ten other unlisted Stage 4 roads, and two below the clause 6.2 minimum were closed pending a stop control. On 28/08/2026, Warrigal technical services added suppressed proximity events to the monthly collision awareness report.

The investigation made three recommendations:

- that BIN-TMP-004 allow intersections to be listed between reviews and require new roads to be referred to the plan's owner
- that CRL-PRO-022 stop excluding earthworks that create a haul road intersection, or that works orders carry the three referrals
- that geofence boundary changes come under change control, with operators told in writing where alerts are suppressed

No safety action had been recorded against the MSG-14 discretion, the contractor induction verification or the night shift extrication arrangement.

Factual information

History of the occurrence

All times are local (UTC +8:00).

The RL 312 dewatering spur was a 240 m access road to dewatering bore DW-14, cut under works order WO-26-4471. Earthworks were complete on 31/07/2026. The spur opened to light vehicles on 02/08/2026, when the bore was energised. That day the "ROAD CLOSED - CONSTRUCTION" sign at the spur mouth was removed and not replaced.

At about 22:10 on 04/08/2026, a dewatering technician drove out of the spur in front of a descending haul truck about 60 m away, then reversed to let it pass. He submitted hazard report HAZ-2026-0917 at 23:48. The Dewatering Superintendent closed it on 06/08/2026.

The light vehicle driver, a grade control technician employed by Merrivale Survey & Geotechnical, received a site access card on 09/08/2026 and worked night shifts on 09/08, 10/08 and 11/08.

At 02:22:41 on 12/08/2026, dispatch called the Stage 4 pit back to work after crib. HT-407, a 220 t class rigid haul truck, began its first post-crib cycle. At 02:38:55 the operator reported "407 dumped, coming back down."

The driver had picked up levels at the DW-14 collar. From the spur length and the vehicle's position at first sighting, LV-118 was estimated to have left the collar at about 02:45 and reached the spur mouth at about 02:47. The driver recalled leaving at about ten to three, later than the recorded collision time. He recalled slowing at the mouth, looking left up the ramp, seeing nothing, and driving out at about 25 km/h without stopping. Neither operator made a positive radio call before entering the intersection.

The truck operator first saw the light vehicle at about 40 m. He braked, applied maximum retard and steered 14 degrees right, towards the windrow. The truck's front left struck the right side of LV-118 at 02:47:12.0, pushing it 18.4 m down the ramp onto its left side against the inner windrow.

The operator called dispatch at 02:47:41. At 02:48:01 dispatch transmitted "All Stage 4 units stop." The operator was told to remain in the cab and was released at 02:48:44. The emergency response team was paged at 02:48. The first response vehicle arrived at 02:54 and the second responder, with the hydraulic rescue tool, at 03:07. The driver was extricated through the rear of the vehicle at 03:14, 27 minutes after the collision. He reached the surface medical facility at 03:41 and was flown to the metropolitan trauma centre at 07:40.

The driver remained conscious. He sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax. LV-118 was destroyed. HT-407 returned to service on 14/08/2026. The Stage 4 pit stood down for 9 hours 15 minutes and the main ramp below RL 340 was closed for 22 hours. The occurrence was notified to the State Mines Inspectorate as a reportable injury.

At 05:10 the Mining Manager directed that, until further notice, a positive call on the pit channel be made before entering any Stage 4 intersection, listed in the plan or not.

Personnel information

Haul truck operator

The operator had 11 years in open pit mining, 7 on 220 t class rigid haul trucks, 14 months at Bindarra and 4 years 2 months with Warrigal Mining Services. His verification of competency on type, issued in 06/2025, was current. He completed an annual refresher in 02/2026 and the site general induction in 06/2025, with no traffic management refresher since. The induction's traffic

management module named the nine Appendix B intersections and did not address unlisted intersections.

He was on night 10 of a 14 night swing, 10 hours 47 minutes into a 12 hour 30 minute shift, and had slept 6 hours 30 minutes in the preceding day. He described himself as tired in the way usual for night ten, but not falling asleep. His drug and alcohol test was negative.

He knew the spur had been cut and had noticed the closed sign was gone, but believed the spur was still closed.

Night shift supervisor

The supervisor had 19 years in open pit mining, 2 years 6 months in the role and 8 years with Warrigal Mining Services. He completed a supervisor refresher in 11/2025. He was on night 10 of 14. His drug and alcohol test was negative.

He was responsible for the haul, excavator, ancillary and dozer fleets and the dispatch operator, not for the dewatering crew, the survey contractor or other light vehicles. He did not know a light vehicle was in the pit that night. He knew the spur had been cut and that the dewatering crew were using it.

Light vehicle driver

The driver had 11 years in survey, 6 in mining and grade control, and 4 years with Merrivale. He was on the third night of his first Bindarra swing, working 18:00 to 06:00, and had driven the spur six or seven times in the preceding week.

He completed the Merrivale general mining induction, a two hour online module, on 24/07/2026, and a 20 minute gatehouse briefing on muster points, speed limits and personal protective equipment. There was no record of the Bindarra site-specific supplement or the Bindarra light vehicle pit driving assessment. He had not heard of the assessment, had not seen the traffic management plan and had not been told to make radio calls in the pit. His radio was on channel 12, as the dewatering foreman had directed. [NOT ESTABLISHED: whether channel 12 was the Stage 4 pit channel]

Aircraft and equipment information

HT-407 had a fleet management system (FMS) and a collision awareness system (CAS). Its monthly brake test on 03/08/2026 recorded 1.9 m/s² deceleration. The FMS in-cab map for Stage 4, last published on 24/07/2026, did not show the spur. The operator described the display at RL 312 as showing the ramp with no intersection.

The CAS operated at Level 7 (operator advisory). Level 9 (automated intervention) was available on the fleet but not enabled at Bindarra. Alerts were raised when a tag entered a 60 m detection ring closing faster than 10 km/h, except inside a suppression geofence. Geofence S4-LOAD was drawn on 14/03/2025 around the Stage 4 loading face. The contractor's technical services group extended it northward on 09/06/2026, to a boundary 180 m up-ramp of the spur intersection. No one on site had a training record on the suppression geofences.

LV-118 carried no FMS and was not dispatched. It had a passive CAS tag and a 4.5 m amber beacon whip. The driver stated that the beacon and lights were on and the vehicle drove normally. [NOT ESTABLISHED: maintenance status of LV-118]

Meteorological information

[NOT ESTABLISHED: recorded weather or visibility conditions]

The truck operator said some dust was hanging and he could not say whether it made a difference. The driver was firm that the air was clear and he could see the ramp surface and windrow.

Recorded data

The FMS download (WMS-TS-2026-114) recorded HT-407 at 41 km/h at RL 368, 39 km/h at RL 348, and 37 km/h at RL 314 at 02:47:06, the last steady point, 41 m above the intersection. At 02:47:07.4, 1.4 seconds later, service brake pressure rose from 0 to 620 kPa in 0.4 seconds. Speed was 32 km/h at 02:47:09.2 and 24 km/h at collision at 02:47:12.0. The truck stopped 19 m below the intersection at 02:47:14.3. Mean deceleration from brake application to collision was 1.85 m/s². The retarder and service brakes performed as specified. Without braking, collision speed would have been 37 km/h.

The CAS log recorded a proximity event at 02:47:09.2, with LV-118 at 21.4 m closing at 13 km/h. Being inside S4-LOAD, the alert was suppressed and nothing appeared in the cab. Outside the geofence it would have raised an advisory tone and screen indication. Impact was logged at 1.9 g at 02:47:12.0.

The pit channel recording (RAD-2026-0332), checked to the FMS clock within 0.4 seconds, contained no positive call at the spur before the collision.

Aerodrome and site information

At RL 312 the main ramp ran approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows, with a dry, well maintained surface. The spur joined from the east on the inside of a left-hand bend at an included angle of 71 degrees, 34 m beyond a crest.

A descending operator at 5.80 m eye height first had line of sight to the spur mouth at 41 m. A driver at the spur mouth at 1.45 m eye height first had line of sight to a descending truck at 38 m. A 1.6 m inner windrow separated spur and ramp for the last 25 m of the spur. Traffic management plan clause 6.2 required at least 120 m sight distance. Calculated stopping distance for an empty 220 t class truck at 40 km/h on that grade was 96 m.

The intersection had no warning sign, give-way or stop control, delineation or posted spur speed.

The truck's left front tyre scuff began 38.5 m up-ramp of the spur mouth, ran 62 m and deviated 1.4 m right over its first 20 m. Impact was 6.2 m into the ramp and 3.1 m from the inner windrow toe. The light vehicle's 18.4 m gouge changed from tyre scuff to body panel at 11.2 m. LV-118 damage was concentrated on the right side from the B pillar forward. The roll-over protection structure was deformed inward at the right rear mounting but not breached. The seatbelt was fastened and its webbing showed loading marks. The beacon whip was sheared and found 9 m away, and its lamp worked when tested. HT-407 damage was confined to the front left bumper, left mudguard and lower two ladder steps.

Organisational and management information

Traffic management plan BIN-TMP-004 required positive calls at nine intersections listed at Appendix B. Clause 6.4 required signs, give-way control and delineation at intersections on the plan's road layout drawing, which did not include the spur. Annual reviews in 11/2023, 11/2024 and 11/2025 reissued Schedule 1 and Appendix B. No out-of-cycle review under clause 9.1 had occurred in the plan's eleven years. Of 14 new Stage 4 roads cut since 12/11/2025, the 11 cut under works orders were not referred to the plan owner, and the 3 cut under a management of change record were.

Works order WO-26-4471 classified the work as "Routine earthworks within the approved Stage 4 design" and stated "Management of change not required, CRL-PRO-022 clause 5.3." Its handover field read "N/A". The geofence extension was not recorded in, or reviewed against, the plan.

All three people interviewed described calling only at the nine listed intersections. The supervisor had never corrected anyone for not calling elsewhere. No permit, call-in or entry check existed for light vehicles entering an operating pit. The supervisor knew from mid 2025 that alerts were suppressed at the dig but not of the 2026 extension. The standing CAS alert report did not include suppressions.

Contractor Management Standard CRL-STD-009 clauses 7.3 to 7.5 permitted a contractor induction to be accepted as equivalent, but stated that the site supplement and pit driving assessment "shall not be substituted" and must be verified by the engaging department before a card is issued. The gatehouse issued cards against the recognition agreement alone.

One mines rescue paramedic was rostered on night shift. A second responder, required before any extrication, was called from the accommodation village 11 km from the gatehouse. The post-incident review found extrication was delayed by about 9 minutes without affecting the clinical outcome.

Additional information

HAZ-2026-0917 described the rise, the windrow, the missing sign and give-way, and stated "I do not think they know the road is there." Rated Medium by the submitter, it was closed at Low with a pre-start reminder to give way to haul trucks and "No further action required." It was not referred beyond the dewatering department.

A search of hazard reports for the 24 months to 11/08/2026 found three others, from 03/2025, 09/2025 and 02/2026, each describing a light vehicle and haul truck meeting unexpectedly at a different unlisted intersection. Each was closed within three days with a crew reminder. None was referred to the plan owner or led to any change.

Analysis

External Influences

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

The regulation under which Bindarra managed vehicle interaction requires only that "the operator shall manage the risks arising from the interaction of vehicles at the mine". Mines Safety Guideline MSG-14, revision 4, is one way of meeting that requirement. Its text is documentary and not in dispute. It is virtually certain that it leaves the discretion described below.

The guideline leaves three terms to the site:

- Clause 3.2 requires a review "following any significant change to the mine's road network, vehicle fleet or operating pattern". It does not define a significant change or give an example of one.
- Clause 5.1 requires the operator to nominate the interaction zones at which positive communication is required. It sets no minimum and no basis for the nomination.
- Clause 5.4 requires a light vehicle to be detectable by the heavy vehicle operator. Any one of five means satisfies it.

Clause 3.5 provides that the plan "is not approved by, and need not be submitted to, the Inspectorate" except on request after an occurrence or an inspection.

Bindarra's plan met the guideline on its face. It had been reviewed within twelve months. It nominated nine intersections at Appendix B. It provided for detectability by line of sight and elevated marking. The mine was not in breach. This factor does not concern compliance, and it does not concern any failure by the Inspectorate. It concerns the shape of the discretion the guideline leaves, and what the site built within it.

At Bindarra all three discretions were filled by one document, BIN-TMP-004. It was reviewed once a year, and nobody outside the mine reviewed any part of it. Clause 9.1 of the plan repeats the guideline's undefined trigger without defining it. An intersection reaches Appendix B only through the road layout drawing at Schedule 1, and that drawing is reissued only at the annual review. No provision allows an intersection to be added between reviews. No provision requires a new road to be referred to the plan's owner when it is cut. At an intersection not on the list, clause 4.3.2 states that "No call is required."

The document register showed how that structure worked in practice:

- No out-of-cycle review had been held in the plan's eleven year history.
- Since the 12/11/2025 review, 14 new roads had been cut in the Stage 4 pit. The 3 cut under a management of change record were referred to the plan's owner. The 11 cut under a works order were not.
- In the 24 months before the occurrence, four hazard reports, including HAZ-2026-0917, described a light vehicle and a haul truck meeting unexpectedly at an intersection not on the list. Each was closed within three days with a reminder to the crew. None reached the plan's owner. None changed Appendix B, Schedule 1 or any physical control.

There is significant evidence against this factor's influence. Nothing in MSG-14 stopped Bindarra from acting. The plan's own clause 9.1 allowed an out-of-cycle review at any time. The guideline does not prevent a mine from listing every intersection it has. On that view, Bindarra did not fail to act because the guideline was vague. It failed to act because nothing inside the mine put the question to the plan's owner, and the guideline is only background.

The investigation considered that argument and did not accept it as a complete explanation. A defined trigger for a significant change, and a stated basis or minimum for nominating interaction zones, would have given the eleven unrefereed roads somewhere to go. Each internal control that failed in this occurrence lacked exactly that: a point at which a new road had to be referred, assessed, and then either listed or controlled. The guideline left that point to the site, and the site left it undefined.

The internal failures are contributing factors in their own right and are argued elsewhere. The opening of the spur under a works order with no management of change is argued under Organisational Influences. The limiting of positive calls to a list rewritten once a year is argued under Risk Controls.

This finding is held with less confidence than the other contributing factors in the occurrence. The argument that the failure inside the mine fully explains the outcome is reasonable, and it was not fully displaced. The evidence favours influence on balance, but the matter is not settled beyond that.

The guideline's open discretion on what counts as a significant change, which interaction zones require positive communication and which means of detectability must be provided likely contributed to a site traffic management plan that had no way to bring a new intersection under its controls between annual reviews.

Organisational Influences

Access spur cut and opened to traffic under a works order, with no management of change raised

CRL-PRO-022 clause 5.2 requires a management of change record before any change to the mine's road network. Clause 5.3 excludes routine earthworks carried out under a works order within an approved pit design where no new plant or vehicle type is introduced. WO-26-4471 cut 240 m of road to bore DW-14 and joined it to the main ramp at RL 312. It classified the work as "Routine earthworks within the approved Stage 4 design" and recorded "Management of change not required, CRL-PRO-022 clause 5.3." The procedure, the works order and the survey department's road register agree with each other. It is virtually certain that the spur was cut and opened to traffic with no management of change raised.

The exclusion mattered because of what the management of change record carries. Clause 5.4 places three referrals inside it:

- the traffic management plan owner
- the survey department, for the fleet management system road network
- the contractor's technical services group, for the collision awareness system configuration

The works order form at Schedule 4 has no equivalent fields. WO-26-4471 carried no signage scope, no survey pickup, no referral and no road network request. Its handover field read "N/A". None of the three referrals was made.

The consequences followed from that gap:

- The in-cab map published on 24/07/2026 was still current on the night. It showed the main ramp with no intersection at RL 312.
- The spur was not on the traffic management plan's road layout drawing. The signage, give-way and delineation requirements of clause 6.4 therefore did not apply to it.
- The clause 6.2 sight distance requirement was never assessed against it.

- When the "ROAD CLOSED - CONSTRUCTION" sign was removed on 02/08/2026, nothing replaced it.
- The intersection was not among those requiring a positive call.

Those matters are argued under Local Conditions and Risk Controls. The point here is narrower. A management of change record was the only route by which the new intersection could reach the plan, the in-cab map or the collision awareness system configuration. Its absence left all three untouched.

This was not a single lapse. Fourteen new roads were cut in Stage 4 after the 12/11/2025 plan review. Three were raised under a management of change record, and all three were referred to the plan's owner. None of the eleven raised under a works order was referred.

One consideration weighs against the influence. A referral is not a control. Had the spur been referred, the plan's owner might have considered it and still left it unlisted. That possibility is why the influence is not put higher. It does not alter the fact that no review of any kind took place.

It is very likely that the absence of a management of change record contributed to the occurrence, because it left the new intersection outside every control that would otherwise have followed it onto an operating haul road.

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

CRL-STD-009 clause 7.3 allows a contractor's own general induction to be accepted as equivalent where the contractor holds a current recognition agreement. Merrivale Survey & Geotechnical has held one since 2023. Clause 7.4 states that equivalence "does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment". It also states that the assessment "shall not be substituted". Clause 7.5 makes the engaging department responsible for verifying both before a site access card is issued.

The standard was sound. The verification it required had nowhere to happen:

- The gatehouse issued cards against the recognition agreement alone.
- It held no record of the supplement or the assessment and was not asked for one.
- Nothing in the access card system distinguished a person who may drive in an operating pit from one who may not.

The driver's card was issued on 09/08/2026 after a 20 minute gatehouse briefing on muster points, speed limits and personal protective equipment. His training file shows the Merrivale online module of 24/07/2026 and no supplement or pit driving assessment. He had never seen BIN-TMP-004 and did not know it existed. The occurrence was on his third night shift at the mine.

It is extremely likely that the driver had completed neither item. The residual doubt is that Merrivale's own records of the site supplement were not examined.

The pit driving assessment is where a light vehicle driver is taught how the pit's traffic rules work in practice. That includes give-way, positive calls and what to do at a blind intersection. Without it, the driver reached the spur mouth with no site instruction on entering a haul road whose approach he could not see. His entry onto the ramp without stopping and without establishing the truck's position is argued under Individual Actions.

Two considerations limit the influence. First, the rule he would have been taught, that light vehicles give way at all times, was one he was already applying. Second, the intersection's geometry defeated attention on its own. From the spur mouth, a descending truck first came into view at 38 m. A driver who had completed the assessment would still have faced that sight line.

The failure lay in the verification arrangements, not in the driver. The standard required a check that the access card process had no means to make.

It is likely that the absence of a verified site supplement and pit driving assessment contributed to the occurrence, because it left the driver without site instruction on entering the ramp at an unlisted, blind intersection.

Risk Controls

Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

It is virtually certain that the suppression existed. The configuration, the event log and the twelve month statistics came from the system itself and were downloaded sealed.

The S4-LOAD geofence was extended northward on 09/06/2026 to a boundary 180 m up-ramp of the RL 312 intersection. The contractor's technical services group made the extension as a system configuration task. It was not recorded in BIN-TMP-004 and was not reviewed against it. No person on site had been trained on where the geofences lay. At 02:47:09.2 the system logged LV-118 at 21.4 m, closing at 13 km/h, and suppressed the alert.

Two readings of influence were considered.

On the first, the missing alert changed nothing. It would have sounded 1.8 seconds after the operator began braking and 2.8 seconds before impact, so it would have told him what he already knew. The system was advisory only, and Level 9 intervention was not enabled. With 41 m of sight line, the outcome was fixed before either the alert or the sighting.

The investigation accepted the second reading, which does not rest on the final seconds. In the twelve months before the occurrence, HT-407 raised 412 advisory alerts, all outside the geofence and none inside it. Across the Stage 4 fleet, 68 proximity events inside the geofence were suppressed. Nine of these involved a light vehicle tag on the main ramp below the boundary, which is the same configuration as this occurrence. Anyone who queried the system could see these events. The technical services group's standing report did not cover suppressed events, however, and nobody had queried it.

An operator who heard alerts everywhere else and never on the ramp had grounds to read silence as an empty road. The truck operator raised this expectation in his own account before he was asked, as did the supervisor, and the statistics are consistent with it. The mechanism is inferred from those accounts and the alert pattern. It was not measured.

The geofence itself was not the deficiency, because suppression around a loading face prevents nuisance alerts that would make the system unusable. The deficiency was that the boundary moved 180 m and nobody outside the group that moved it was told.

It is likely that the suppression contributed by giving the truck operator an inferred expectation that the ramp was clear, although the absent alert itself would have come after he had begun braking.

Positive radio calls required only at listed intersections, and the list is rewritten once a year

Clause 4.3.1 of BIN-TMP-004 requires a positive call before entering "any intersection listed at Appendix B". Clause 4.3.2 provides for all other intersections that "No call is required." An intersection reaches Appendix B only where it appears on the Schedule 1 drawing and the annual review assesses it. Schedule 1 is reissued only at that review. No provision allows an addition

between reviews, and none requires a new road to be referred to the plan's owner. The RL 312 spur was on neither document. It is virtually certain that the control was arranged this way, because the plan, its appendix and its review history are documentary.

The arrangement operated as written. Fourteen roads were cut in Stage 4 after the 12/11/2025 review, and the 11 cut under works orders were never referred. No out-of-cycle review had been held in the plan's eleven year history. Earlier hazard reports from 03/2025, 09/2025 and 02/2026 each described a light vehicle meeting a haul truck at an unlisted intersection. HAZ-2026-0917 described the same at RL 312 seven days before the collision. All four were closed with a reminder to the crew, and none reached the plan's owner.

The truck operator, the light vehicle driver and the night shift supervisor each described the same practice independently. Calls were made at the nine intersections on the crib room board and nowhere else. The truck operator's induction named those nine and said nothing about other intersections. The earliest date on which the spur could have been listed was 12/11/2026.

The pit channel recording shows no call from either vehicle about the spur, the ramp or a light vehicle before the collision. Had a call been required, the conditions for it to work were present. The channel was open and both vehicles were on it. Dispatch was tracking HT-407 closely enough at 02:46:18 to place it "about four minutes off" the loading face. The residual doubt is that a call made and not acknowledged would still leave a driver deciding whether to go.

The opening of the spur without referral and the guidance that left the review trigger to the site are argued under Organisational Influences and External Influences. Neither operator's omission of a call is argued here.

It is very likely that limiting positive calls to an annually reissued list contributed, because it left the RL 312 intersection without the mine's own defence against this interaction.

One emergency response paramedic rostered on night shift, with the second responder called from the village

The night shift was rostered with one qualified mines rescue paramedic, based at the surface medical facility. The second responder must be present before any extrication begins and is called in from the accommodation village 11 km from the gatehouse. It is virtually certain that this arrangement existed, because the roster, the response log and the post-incident review agree on the times.

Dispatch paged the team at 02:48. The first response vehicle reached the scene at 02:54. The second responder arrived with the hydraulic rescue tool at 03:07, and the driver was extricated at 03:14, 27 minutes after the collision. The post-incident review found that the extrication was delayed by approximately 9 minutes. It found no effect on the clinical outcome, because the injuries were stable and the driver was conscious throughout.

The collision, the injuries and their severity were complete before the team was paged. It is unlikely that the arrangement influenced the occurrence or its outcome.

The arrangement still matters. The driver was conscious and restrained, and the roll-over protection structure was not breached. A fuel fire, an unconscious casualty or an airway needing access within the first ten minutes would make the same nine minutes a different outcome. The arrangement is the same on every night shift and does not change with the severity of the casualty.

It is very likely that the night shift extrication arrangement increased risk, although it did not influence this occurrence.

Local Conditions

The spur meets the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m

It is virtually certain that the available sight distance at the RL 312 intersection was 41 m. The mine survey department measured it to the eye heights of both vehicles. A descending operator at an eye height of 5.80 m first had line of sight to the spur mouth at 41 m. A driver at the spur mouth first had line of sight to a descending truck at 38 m, and the 1.6 m inner windrow hid the truck's lower body for the last 25 m of the spur's approach. The dewatering technician's hazard report of 04/08/2026 described the same rise, windrow and blind exit seven days earlier.

The figure that matters is the stopping distance. An empty 220 t class truck at the posted 40 km/h on the 9 per cent descent needs 96 m to stop, allowing a 1.5 second response. Clause 6.2 of BIN-TMP-004 requires sight distance of not less than that figure, and in no case less than 120 m. Where that cannot be achieved, it requires a stop, signalised or controlled intersection. The intersection met neither limb. At 41 m, a truck at the posted speed could not stop for anything at the spur mouth, however alert its operator. The geometry left about four seconds between first sight and the intersection.

The recorded data bears this out. The operator of HT-407 responded within 1.4 seconds, and the truck still reached the intersection at 24 km/h. Attention, training and a radio call can only act within the margin a road provides, and here the geometry had already removed it. The intersection also carried no warning sign, give-way control, delineation or posted speed, because clause 6.4 applies only to intersections on the plan's road layout drawing. This intersection was not on that drawing. How the spur was opened in this condition is argued under Organisational Influences.

The sight line did not by itself put the light vehicle in the truck's path. That also required the absence of positive radio calls, the suppressed collision awareness alert and the light vehicle's entry without stopping. Those matters are argued in their own subsections.

The 41 m sight line, less than half the 96 m stopping distance, very likely contributed to both vehicles arriving at the intersection with neither operator aware of the other.

Dust on the ramp reduced the truck operator's ability to see the light vehicle

The investigation examined whether dust reduced what the truck operator could see. It could not settle the question. Dust affecting sight was found about as likely as not.

Some evidence supported it. The water cart last passed that section at 01:10, 1 hour 37 minutes before the collision. The standing interval is 90 minutes, so a pass was 7 minutes overdue. The fixed dust monitor at the ramp crest, 180 m up-ramp of the intersection, logged PM10 at 142 micrograms per cubic metre between 02:40 and 02:50, against a shift mean of 88. The truck operator described dust hanging on the ramp. He was also descending behind his own lights into a pit with no fixed lighting, which is the condition in which airborne dust is worst for a driver.

The evidence against was stronger in one respect. The light vehicle driver was at the same place about four seconds earlier. He was firm that the air was clear and that he could see the ramp surface and the windrow. The truck operator declined to say that dust made any difference. The reading of 142 was well below the site trigger of 250. It measured particulate at a monitor head 180 m away, not visibility from either cab. Nothing at Bindarra measures visibility. Most significantly, the operator sighted the light vehicle at 41 m, which is the distance the survey gives for clear air. The sighting is fully explained by the geometry, so dust has nothing left to explain.

The deviation in watering and particulate was real, and it occurred at the right place and time. Its effect on what either driver could see was never measured. It cannot be reconstructed from the

evidence that exists. Settling the question would require a visibility measurement from the cab on the night, and no such measurement exists.

Whether dust reduced the truck operator's ability to see the light vehicle remains unresolved, and no conclusion is drawn from it.

The ramp surface was wet and the truck could not stop in the distance available

It is very unlikely that the ramp was wet. The met station recorded no precipitation, the last rainfall was 26 days earlier and the investigator found the surface dry at 06:20. The truck also achieved a mean deceleration of 1.85 m/s², against its tested dry figure of 1.9 m/s², with no sign of a locked or sliding wheel.

Night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low

It is virtually certain that the truck operator was on night 10 of a 14 night swing. The collision occurred 10 hours 47 minutes into an 18:00 to 06:30 shift, within the window of lowest circadian alertness. Every limit in the contractor's fatigue management standard was met. That is where degraded detection and slowed response would be expected.

The recorded data does not show either effect. The operator's response time was 1.4 seconds. That is inside the 1.5 seconds assumed in the plan's stopping distance calculation and faster than the fleet figure. His braking and steering were immediate and in the only available direction. At 41 m and 37 km/h, a collision would have occurred even with no response delay at all. There was no slowed response or omission for fatigue to act through. It is unlikely that the operator's fatigue state influenced the occurrence.

The exposure still matters. The Stage 4 night roster puts every haul truck operator in the small hours of a tenth consecutive night, in a pit with no fixed lighting. At an unlisted intersection, the operator's own eyes are the sole defence. The same roster is worked on three other pits on the site.

Extended night rostering through the circadian low likely increased risk across the haul truck fleet, although it did not contribute to this occurrence.

Individual Actions

Neither operator made a positive call on the pit channel before entering the intersection

It is virtually certain that neither operator made a call on the pit channel about the RL 312 spur before the collision. The channel was logged continuously, and the investigator listened to the whole period from the return from crib to two minutes after the collision. No transmission by either vehicle referred to the spur, the bore, or a light vehicle in the Stage 4 pit. LV-118 made no transmission that shift after logging on at 18:04. The truck operator, the light vehicle driver and the night shift supervisor were each interviewed separately, and each confirmed this.

The omission was consistent with the plan. Clause 4.3.2 of BIN-TMP-004 states: "At intersections not listed at Appendix B, vehicles shall give way in accordance with clause 4.1. No call is required." The RL 312 intersection was not listed. All three interviewees independently described the same practice. Calls were made at the nine intersections on the crib room board and nowhere else. The supervisor stated that he had never corrected anyone for not calling elsewhere, and that requiring calls everywhere would make the channel unusable. The operators were following the plan and the established practice on the pit.

A call from either vehicle would have been heard by the other vehicle, by dispatch and by the supervisor. Dispatch was already tracking the truck closely enough to tell the excavator operator at 02:46:18 that "407's about four minutes off you". A call would very probably have placed each vehicle in the other operator's knowledge. However, a call provides knowledge and does not provide separation. With 41 m of sight line, the two vehicles would still have had to establish which went first. For that reason the contribution is assessed below certainty.

The organisational reasons the intersection carried no call requirement are argued under Risk Controls.

It is likely that the absence of a positive call contributed to both vehicles arriving at the intersection with neither operator aware of the other.

Light vehicle entered the ramp from the spur without stopping and without establishing the truck's position

It is virtually certain that the light vehicle entered the ramp without stopping. On read-back the driver corrected his own account from stopped to slowed, which was against his own interest. His corrected account was that he left the spur at about 25 km/h, looked left up the ramp, saw nothing and drove out. The scene evidence agrees with this account. The impact point was 6.2 m into the ramp from the eastern edge of the spur mouth.

The driver described himself as applying clause 4.1, which requires light vehicles to give way to heavy vehicles at all times. Nothing at the mouth indicated that the intersection required more than this. There was no give-way control, stop line, warning sign or posted speed. Clause 6.4 applies only to intersections on the road layout drawing, and the spur was not on it. The driver had worked three night shifts at Bindarra, had done no Bindarra pit driving assessment, and had never seen the traffic management plan.

The entry was the step that placed the light vehicle across the path of a descending haul truck. Without it there would have been no collision, and it was the last point at which a different decision would have avoided one. The survey limits how much weight this can carry. From the mouth, at the driver's eye height, the sight line was 38 m. The windrow also hid the truck's lower body for the last 25 m of the approach. A full stop would therefore have shown him about what he saw at 25 km/h. It would have added a few metres of sight line, against the 96 m the truck needed to stop. The sight distance is argued under Local Conditions. The induction that left the driver without the site supplement and pit driving assessment is argued under Organisational Influences.

It is very likely that the light vehicle's entry onto the ramp, without stopping and without establishing the truck's position, contributed to both vehicles arriving at the intersection with neither operator aware of the other.

Truck operator braked and steered into the windrow within 1.4 seconds of sighting

It is virtually certain that the truck operator responded within 1.4 seconds. The fleet management system download, corroborated by the scene marks, recorded the last steady data point at 02:47:06, at 37 km/h and 41 m above the intersection. Service brake pressure began to rise 1.4 seconds later and reached 620 kPa within 0.4 seconds. The retarder was taken to maximum and 14 degrees of right steering was applied at the same time. This response was within the 1.5 seconds assumed in the plan's own stopping distance calculation.

The braking reduced the impact speed from 37 km/h to 24 km/h. The mean deceleration was 1.85 m/s², against the unit's tested 1.9 m/s². The braked scuff deviated 1.4 m to the right over its first 20 m, towards the wall side of the road. This kept the light vehicle against the inner windrow, away

from a drop of about 100 m on the other side. The light vehicle's restraint and roll-over protection then held.

After stopping 19 m below the intersection, the operator reported the collision at 02:47:41. He followed the dispatcher's instruction to stay in the cab until the ramp was stopped, and then stayed with the driver until the response team arrived.

It is very likely that the truck operator's braking and steering reduced the severity of the outcome, keeping the light vehicle on the wall side of the ramp and lowering the impact speed from 37 km/h to 24 km/h.

Occurrence Events

Both vehicles arrived at the intersection with neither operator aware of the other

At 02:47:06, HT-407 was descending the main ramp empty at 37 km/h, 41 m above the RL 312 intersection. LV-118 was leaving the spur at about 25 km/h, with 38 m of sight line up the ramp.

Neither operator had any means of learning of the other before that point:

- The truck's in-cab road network showed no intersection at RL 312.
- There was no warning sign on the ramp approach and no give-way control at the spur mouth.
- No call was made on the pit channel.
- The CAS on HT-407 detected the light vehicle and logged "range 21.4 m closing 13 km/h". It recorded the alert as "SUPPRESSED (geofence)", and nothing was presented in the cab.
- The night shift supervisor did not know a light vehicle was in the pit. There was no permit, call-in or board by which he could have known.

The first sighting available to either operator was at 41 m, about four seconds before the vehicles met. This is what the survey predicts for the geometry. The pit channel recording, the FMS download, the CAS log and all three interviews were examined. None of them contained any indication that either operator knew of the other before that moment.

Three independent recordings and all three interviews agree. The truck's movements and the final two seconds of the light vehicle's approach are fixed by recorded data. The light vehicle's movements before those final two seconds rest on the driver's account alone. That is the only reason this condition is not stated more strongly.

The condition was the point at which the separate lines of enquiry met. These were:

- the suppressed CAS alerts, argued under Risk Controls
- the 41 m sight line against a plan minimum of 120 m, argued under Local Conditions
- the absence of calls on the pit channel, argued under Individual Actions
- the light vehicle's entry onto the ramp without stopping, argued under Individual Actions.

The analysis treats them as one mechanism rather than as separate causes.

Whether dust on the ramp reduced the truck operator's ability to see the light vehicle was examined and could not be resolved. That matter is discussed under Local Conditions, and nothing from it is carried into this conclusion.

Once the vehicles had converged, the collision followed directly. The survey gives a stopping distance of 96 m for the truck at the posted speed. Nothing between the convergence and the impact was capable of preventing the collision at 41 m.

It was extremely likely that both vehicles arrived at the intersection with neither operator aware of the other, and extremely likely that this contributed to the collision.

The haul truck's front left struck the right side of the light vehicle at 24 km/h

The recorded data, the damage pattern and the debris field agree.

- **Timing and speed.** The FMS download places the impact at 02:47:12.0 at 24 km/h, 4.6 seconds after the brake application. The truck's accelerometer logged "longitudinal 1.9 g" at the same time.
- **Position.** The scene examination placed the impact 6.2 m into the ramp from the eastern edge of the spur mouth, 3.1 m out from the inner windrow toe.
- **Damage.** Damage to LV-118 was concentrated on its right side from the B pillar forward. Damage to HT-407 was confined to the front left bumper, the left mudguard and the lower two steps of the access ladder, with nothing above the bumper line.
- **Debris.** Right-side glass and mirror fragments lay from 4.8 m to 14 m down-ramp of the impact position.

The spur joins the ramp from the east, on the inside of a left-hand bend, at an included angle of 71 degrees. A vehicle leaving the spur was therefore crossing the path of a descending truck. The truck struck the side of the light vehicle, not its front. That side impact is what the roll-over protection then had to withstand.

The truck operator's braking reduced the impact speed from the 37 km/h that would otherwise have occurred to 24 km/h. That response is argued under Individual Actions. With only 41 m available, the outcome then depended on the light vehicle's structure, its restraint and the impact speed.

It was virtually certain that the haul truck's front left struck the right side of the light vehicle at 24 km/h, and virtually certain that this contributed to the occurrence.

The light vehicle was pushed 18 m down the ramp and rolled onto its left side

The scene measurements, the vehicle examination and the medical evidence agree.

- **Path.** A continuous gouge ran 18.4 m down-ramp from the impact position. At 11.2 m it changed from tyre scuff to body panel gouge, which marks where the vehicle went over.
- **Rest position.** LV-118 came to rest on its left side against the inner windrow, facing down-ramp.
- **Injuries.** The driver sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, and was conscious throughout.
- **Extrication.** The door on the resting side was against the ground, so extrication was through the rear of the vehicle.

The roll-over was the means by which the collision produced these injuries. The injuries are consistent with a side impact and roll-over in which the structure held and the restraint was worn.

Two protective measures worked:

- The roll-over protection structure was deformed inward at the right rear mounting but was not breached.
- The driver's seatbelt was found fastened, and the webbing showed loading marks.

Without either of these, the same sequence would very likely have had a different outcome. The emergency response that followed is discussed under Risk Controls.

It was virtually certain that the light vehicle was pushed 18 m down the ramp and rolled onto its left side, and virtually certain that this contributed to the driver's injuries.

The haul truck's service brakes were degraded and could not achieve the required stopping performance

It was very unlikely that HT-407's service brakes were degraded, because brake tests on 03/08/2026 and 14/08/2026 gave mean decelerations of 1.9 and 1.92 m/s², the occurrence download gave 1.85 m/s² between them, the pre-start check for 11/08/2026 recorded no defects, and no brake, retarder or park brake defect had been raised in the preceding twelve months.

Conclusions

These findings are made with respect to the occurrence and should not be read as apportioning blame or liability to any organisation or individual.

Findings

1. It is virtually certain that Mines Safety Guideline MSG-14 required a review of the traffic management plan after any significant change to the road network without defining one, and left the nomination of positive communication intersections to the site with no minimum and no basis. It is likely that this discretion contributed, by giving roads cut between annual reviews no defined route to the plan. The mine was not in breach. The contrary reading, that Bindarra could have held an out-of-cycle review or listed every intersection at any time and that the failure lay wholly inside the mine, was not excluded. (Safety issue)
2. It is virtually certain that the RL 312 dewatering spur was cut and opened to traffic under works order WO-26-4471 with no management of change raised, because CRL-PRO-022 clause 5.3 excluded routine earthworks. It is very likely that this contributed, because none of the three referrals a management of change record would have triggered was made, and the traffic management plan, the FMS in-cab road network and the CAS geofence configuration were all left unchanged. (Safety issue)
3. It is very likely that the light vehicle driver's contractor induction was accepted as equivalent without the Bindarra site-specific supplement or the Bindarra light vehicle pit driving assessment ever being verified, although CRL-STD-009 stated that neither shall be substituted. The access card system held no field for either. It is likely that this contributed to the light vehicle entering the ramp without establishing the truck's position. The give-way rule the assessment teaches was one the driver was already applying, and the sight line limited what attention alone could achieve. (Safety issue)
4. It is virtually certain that the S4-LOAD suppression geofence, extended northward on 09/06/2026 without being recorded in or reviewed against the traffic management plan, placed the RL 312 intersection inside the area where CAS alerts were suppressed. It is likely that this contributed through the truck operator's expectation, built over his time at the mine, that the absence of alerts on that ramp meant an empty road, rather than through the single alert suppressed at 02:47:09.2. That alert would have come 1.8 seconds after braking began, and the reading that the suppression did not cause this collision was not excluded. (Safety issue)
5. BIN-TMP-004 required a positive radio call only at the nine intersections listed at Appendix B. The list was reissued only at the annual review, and no provision required a new road to be referred to the plan's owner, so the RL 312 spur was not listed. This is virtually certain, and it is very likely that it contributed by leaving no call required at the intersection. (Safety issue)
6. It is virtually certain that one mines rescue paramedic was rostered on night shift, with the second responder, who was required before any extrication, called in from the accommodation village 11 km from the gatehouse. Extrication was delayed by about 9 minutes as a result. This did not contribute to the occurrence or to its clinical outcome. It is very likely that it increased risk, because the same delay applies on every night shift whatever the casualty's condition. (Safety issue)
7. It is virtually certain that the spur joined the main ramp 34 m beyond a crest, giving a descending truck operator 41 m of sight line against a plan minimum of 120 m and a stopping distance of 96 m, with no warning sign, give-way control, delineation or posted speed. It is very likely that this contributed, because a truck at the posted speed could not stop for anything at the intersection.

8. The hypothesis that the ramp surface was wet and the truck could not stop in the distance available was examined and ruled out. The surface was found dry, the last rainfall was 26 days earlier, and the deceleration achieved matched the truck's dry tested figure.

9. With virtual certainty, the truck operator was on night 10 of a 14 night swing, 10 hours 47 minutes into a 12 hour 30 minute shift, in the circadian low. It is likely that this standing roster exposure increased risk. It did not contribute to this occurrence, because his recorded 1.4 second response was inside the plan's 1.5 second allowance, and at 41 m of sight line a response time of zero would still have ended in a collision.

10. It is virtually certain that neither operator made a positive call on the pit channel before entering the intersection. Neither the plan nor local practice required a call there. It is likely that this contributed, because a call would have put each vehicle in the other's knowledge, although knowledge alone would not have given separation over 41 m of sight line.

11. The light vehicle entered the ramp from the spur at about 25 km/h without stopping and without establishing the truck's position, which is virtually certain. It is very likely that this contributed, because it placed the light vehicle across the descending truck's path, although a full stop at the mouth would have shown the driver about what he saw on the move.

12. The truck operator braked, took the retarder to maximum and steered 14 degrees right towards the inner windrow 1.4 seconds after the last steady data point. This reduced the impact speed from 37 km/h to 24 km/h and kept the collision on the wall side of the road.

13. It is very likely that HT-407 and LV-118 arrived at the RL 312 intersection with neither operator aware of the other until about four seconds before the collision. It is very likely that this contributed, because the collision followed directly from that convergence.

14. With virtual certainty, the front left of HT-407 struck the right side of LV-118 at 24 km/h at 02:47:12.0, which with virtual certainty contributed to the injuries.

15. With virtual certainty, LV-118 was pushed 18.4 m down the ramp and rolled onto its left side against the inner windrow, which with virtual certainty produced the driver's injuries. The fastened seatbelt was loaded, and the roll-over protection structure deformed without being breached.

16. The hypothesis that the haul truck's service brakes were degraded and could not achieve the required stopping performance was examined and ruled out. Brake tests on 03/08/2026 and 14/08/2026 and the occurrence download all agreed within normal scatter.

Probable cause

The probable cause of this occurrence was that the RL 312 dewatering spur was opened to traffic under a works order with no management of change raised. This left an intersection with 41 m of sight line against a 96 m stopping distance unsigned, unlisted for positive radio calls and absent from the in-cab road network and the CAS configuration, so that the light vehicle entered the main ramp without stopping or establishing the position of a descending haul truck, with neither operator aware of the other until about four seconds before the collision.

A finding stated without a likelihood term was established as fact.

Contributing factors

- The collision awareness alerts were suppressed inside a geofence extended to cover the new intersection, which is virtually certain, and this likely contributed through the truck operator's expectation of an empty road rather than through the suppressed alert itself. (Safety issue)

- Traffic management guidance left the review trigger undefined and the interaction zones to the site, which is virtually certain. That this contributed is likely, and the reading that the failure lay wholly inside the mine was not excluded. (Safety issue)
- It is virtually certain that the access spur was cut and opened to traffic under a works order with no management of change raised, and very likely that this contributed. (Safety issue)
- The spur met the ramp beyond a crest, giving 41 m of sight line against a plan minimum of 120 m. This is virtually certain and very likely contributed.
- Positive radio calls were required only at listed intersections, and the list was rewritten once a year, which is virtually certain. It is very likely that this contributed. (Safety issue)
- It is virtually certain that neither operator made a positive call on the pit channel before entering the intersection, and likely that this contributed.
- The contractor induction was very likely accepted as equivalent with the site supplement and pit driving assessment never verified, and this likely contributed. (Safety issue)
- With virtual certainty, the light vehicle entered the ramp from the spur without stopping and without establishing the truck's position, and this very likely contributed.
- Both vehicles very likely arrived at the intersection with neither operator aware of the other, and this very likely contributed.
- The haul truck's front left struck the right side of the light vehicle at 24 km/h, which is virtually certain and with virtual certainty contributed.
- With virtual certainty, the light vehicle was pushed 18.4 m down the ramp and rolled onto its left side, and this with virtual certainty contributed.

Other factors that increased risk

These factors did not contribute to this occurrence.

- It is virtually certain that one emergency response paramedic was rostered on night shift with the second responder called from the village, and very likely that this increased risk. (Safety issue)
- The truck operator was, with virtual certainty, on night 10 of 14, in the eleventh hour of a 12.5 hour night shift, in the circadian low, and it is likely that this increased risk.

Other findings

- The hypothesis that the ramp surface was wet and the truck could not stop in the distance available was examined and ruled out.
- The hypothesis that the haul truck's service brakes were degraded and could not achieve the required stopping performance was examined and ruled out.
- The truck operator braked and steered into the windrow within 1.4 seconds of the last steady data point, which reduced risk.

The investigation could not determine whether dust on the ramp reduced the truck operator's ability to see the light vehicle. Several points supported it. The last water cart pass on that section was 7 minutes overdue against a 90 minute interval. The fixed monitor at the ramp crest logged PM10 at 142 micrograms per cubic metre against a shift mean of 88. The truck operator described dust hanging on the ramp while descending behind his own lights into an unlit pit. Several points weighed against it. The light vehicle driver was firm that the air was clear. The truck operator could not say whether the dust made a difference. The reading was below the site trigger of 250 and was taken 180 m from the intersection. The first sighting at 41 m matched what the geometry gave in

clear air. Nothing at Bindarra measured visibility, and without a measurement of what could be seen from either cab at the time, the matter could not be settled.

Recommendations

Traffic management guidance leaves the review trigger undefined and the interaction zones to the site

Mines Safety Guideline MSG-14 revision 4, made under Mining Operations Safety Regulations 2018 r 6.22, leaves three terms for the mine to define:

- Clause 3.2 requires the traffic management plan to be reviewed after any significant change to the road network, but does not define such a change or give an example.
- Clause 5.1 requires the operator to nominate the interaction zones at which positive communication is required, with no minimum and no basis for the nomination.
- Clause 5.4 accepts any one of five means of making a light vehicle detectable to a heavy vehicle operator.

Under clause 3.5, the plan is a site document. It is not approved by the Inspectorate and is not submitted to it except on request after an occurrence. The guideline's text is documentary, and the existence of this discretion was virtually certain.

Bindarra's plan met the guideline on its face, and the mine was not in breach. The issue is the breadth of the discretion the guideline leaves, not any failure by the Inspectorate.

The investigation found that this discretion likely contributed to the occurrence. That finding is less firm than the other contributing factors.

Against it, the guideline did not prevent the mine from holding an out-of-cycle review or listing every intersection. The plan's own clause 9.1 allowed an out-of-cycle review at any time.

For it, all three terms were filled in by one site document, reviewed once a year, with no outside review. Eleven of the fourteen new Stage 4 roads were never referred to the plan's owner. A defined trigger and a stated minimum would have given those roads a route into the plan.

No safety action has been recorded against this issue, and no recommendation is made.

Access spur cut and opened to traffic under a works order, with no management of change raised

CRL-PRO-022 requires a management of change record before any change to the mine's road network. Clause 5.3 excludes routine earthworks carried out under a works order within an approved pit design where no new plant is introduced. WO-26-4471 relied on that exclusion to cut 240 m of access to bore DW-14 at RL 312, which created a new intersection with the main ramp. No management of change record was raised. As a result, none of the three referrals at clause 5.4 was made:

- to the traffic management plan owner
- to the survey department, for the fleet management system road network
- to the contractor's technical services group, for the collision awareness system geofences

The absence of the record very likely contributed to the occurrence.

The issue is not confined to this spur. Fourteen new roads were cut in Stage 4 since the 12/11/2025 plan review. The three raised under a management of change record were referred to the plan's owner. None of the eleven raised under a works order was. Each road built this way can enter an operating pit without reaching the plan, the in-cab map or the geofence configuration.

The investigation recommends that [NOT ESTABLISHED: organisation responsible for CRL-PRO-022, as addressee] do one of two things:

- amend CRL-PRO-022 clause 5.3 so that it no longer excludes earthworks that create a new intersection with a haul road, or
- amend the works order form at Schedule 4 so that it carries the three referrals held at clause 5.4.

A response was due by 30/09/2026. [NOT ESTABLISHED: response from the addressee and any safety action taken against this issue]

Contractor induction accepted as equivalent, with the site supplement and pit driving assessment never verified

CRL-STD-009 allows a contractor's general induction to be accepted as equivalent where a recognition agreement is held. It excludes the Bindarra site-specific supplement and the Bindarra light vehicle pit driving assessment from that equivalence, and it makes the engaging department responsible for verifying both before a card is issued. The standard said the right thing. The gatehouse was the only point at which the verification could have taken place. It issued access cards against the recognition agreement alone, held no record of either item and was not asked for one. The access card system did not distinguish a person permitted to drive in an operating pit from one who was not. It was extremely likely that the light vehicle driver had completed neither the supplement nor the assessment. He had never seen BIN-TMP-004.

The pit driving assessment is where a light vehicle driver is taught how the pit's traffic rules work in practice, including give-way, positive calls and conduct at a blind intersection. On his third night at the mine, the driver was operating on an active haul road without the assessment the site's standard says may not be substituted. The absence of verification likely contributed to the light vehicle entering the ramp from the spur without stopping and without establishing the truck's position. That contribution was limited because the driver was already applying the give-way rule the assessment would have taught, and because the 41 m sight line defeated attention on its own.

No safety action has been recorded against this issue.

Positive radio calls required only at listed intersections, and the list is rewritten once a year

The traffic management plan, BIN-TMP-004, required a positive call on the pit channel only before entering one of the nine intersections listed at Appendix B. An intersection reached that list only through the road layout drawing at Schedule 1, which was reissued once a year at the annual review. No provision allowed an intersection to be added between reviews. None required a new road to be referred to the plan's owner when it was cut. The RL 312 spur was on neither document, so the plan's own terms left the mine's defence against this interaction unarmed at this intersection. The earliest date on which the spur could have been listed was 12/11/2026, three months after the collision. Eleven of the fourteen new Stage 4 roads cut since the last review had not been referred.

Safety action

At 05:10 on 12/08/2026, two hours twenty-three minutes after the collision, the Mining Manager issued a site-wide direction. It required a positive call on the pit channel before entering any intersection in the Stage 4 pit, listed in the plan or not, until further notice. The direction was read at the following four pre-starts and posted beside the Appendix B board in the Stage 4 crib room. It stands until BIN-TMP-004 is amended. This action addresses the issue in part. It covers the Stage

4 pit, but it does not change how an intersection reaches Appendix B or require new roads to be referred.

Recommendation

The operator should amend BIN-TMP-004 to provide a route by which an intersection is added to Appendix B between annual reviews. The amended plan should also require a new road to be referred to the plan's owner when it is opened to traffic.

Collision awareness alerts suppressed inside a geofence extended to cover the new intersection

The collision awareness system on the Stage 4 haul fleet suppressed advisory alerts inside the S4-LOAD geofence. That geofence was drawn around the loading face to stop nuisance alerts. On 09/06/2026 its boundary was extended northward to a point 180 m up-ramp of the RL 312 intersection. The extension was made as a system configuration task by the contractor's technical services group. It was not recorded in the traffic management plan or reviewed against it, and no person on site had been trained on the geofences. The suppression likely contributed to the collision, through the expectation that silence on that ramp meant an empty road.

The issue matters beyond this occurrence. HT-407 alone recorded 68 suppressed proximity events inside the geofence in the preceding twelve months, nine of them a light vehicle on the main ramp below the boundary. None was reported to anyone.

On 28/08/2026, Warrigal technical services added suppressed proximity events to the monthly collision awareness report, broken down by geofence and by vehicle class. It also re-cut the previous twelve months on the same basis for the Stage 4 fleet. The first report in this form was due on 30/09/2026. This action addresses the issue in part. It makes suppressions visible, but it does not bring boundary changes under change control or tell operators where alerts are suppressed.

The investigation recommends that [NOT ESTABLISHED: organisation to which the recommendation is addressed] treat a change to a suppression geofence boundary as a change to a documented control under CRL-PRO-022 clause 5.2. It also recommends that operators be told in writing where the collision awareness system is suppressed and why.

One emergency response paramedic rostered on night shift, with the second responder called from the village

The Bindarra night shift was rostered with one qualified mines rescue paramedic, based at the surface medical facility. The second responder had to be present before any extrication began. That responder was called in with the hydraulic rescue tool from the accommodation village, 11 km from the gatehouse.

On the night of the occurrence, dispatch paged the team at 02:48. The first response vehicle entered the pit at 02:51 and reached the scene at 02:54. The second responder arrived with the tool at 03:07. The driver was extricated at 03:14, 27 minutes after the collision. The site's post-incident review found that extrication was delayed by approximately 9 minutes while the team waited for the second responder and the tool.

The arrangement did not influence this occurrence or its outcome. The collision, the injuries and their severity were complete before the team was paged. The post-incident review found on clinical grounds that the delay did not affect the outcome, because the driver's injuries were stable and he was conscious throughout.

It was very likely an important safety issue regardless. On every night shift, the site's extrication capability was a single person plus a drive from the village. No part of the arrangement changed with the severity of the casualty. In a fuel fire, with an unconscious casualty, or with an airway that needed the vehicle opened in the first ten minutes, the same wait could produce a different outcome.

No safety action has been recorded against this safety issue.

Distribution

Title	Name
General Manager, Bindarra Gold Mine	T. Brandner
Registered Manager (statutory)	S. Kirkhope
Mining Manager and owner of BIN-TMP-004	H. Quaile
Manager Health and Safety, Corella Resources	L. Farquhar
Project Manager, Warrigal Mining Services	D. Mokoena
Superintendent Dewatering, Corella Resources	P. Nyaguse